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Potential (intrinsic energy of inertia)

Electric endogen energy $E_{pot} = mc^2$:

$$e = \text{electron charge} \qquad q = ne = \text{charge} \qquad n \in \mathbb{N} \tag{1}$$

$$\text{Potential} \qquad U = \frac{E_{pot}}{q} = \frac{m}{n}\frac{c^2}{e} = \frac{m}{q}c^2 = \kappa c^2 \tag{2}$$

Electric induced energy $E_{kin} = \frac{1}{2}mv^2$:

E_{kin} Energy density electric:

$$w_e = \frac{1}{2}\epsilon_0 \mathcal{E}^2 \tag{3}$$

E_{kin} Energy density magnetic:

$$w_m = \frac{1}{2}\frac{1}{\mu_0}\mathcal{B}^2 \tag{4}$$

E_{kin} Electric Energy:

$$W_e = \frac{1}{2}CU^2 \tag{5}$$

E_{kin} Magnetic Energy:

$$W_m = \frac{1}{2}LI^2 \tag{6}$$

Power:

$$P = VI = (U_1 - U_2)\frac{Q}{t} \equiv \frac{\text{Work}}{\text{Time}} = \text{Force} \cdot \text{Velocity} = \frac{\text{Energydifference}}{\text{Time}} \tag{7}$$

Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \epsilon_0 \tag{8}$$

Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c}\frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \tag{9}$$

Energy conservation for (large extended) conductive medium:

$$\begin{aligned}\int\left(\frac{\partial}{\partial t}\left(we+w_m\right)+\;\cdot\mathcal{S}+j\cdot\mathcal{E}\right)\,dr&=\frac{d}{dt}W_e+\frac{d}{dt}W_m+D_S+\int\left(j\cdot\mathcal{E}\right)\,dr\\ &\overset{lowfrequency}{=}\frac{1}{2}\frac{d}{dt}\left(LI^2\right)+\frac{1}{2}\frac{d}{dt}\left(\frac{Q^2}{C}\right)+\left(RI^2-VI\right)=0\end{aligned}\tag{10}$$

Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potentialdifference, tension):

$$V=U_1-U_2=r\mathcal{E}=\dot{\Psi}+RI+\frac{I}{C}t=L\dot{I}+RI+\frac{Q}{C}\tag{11}$$

$$V=\text{Potentialdifference}\quad I=\text{Current}\quad R=\text{Resistance}\quad j=\text{Current density}$$

$$L=\text{Inductance}\qquad C=\text{Capacitance}\qquad Q=\text{charge (ion)}\qquad \Psi=LI=\text{mag. Flux}$$

$$\mathcal{S}=\mathcal{E}\times\mathcal{H}=\text{Poynting-Vector}=\text{radiation intensity}$$

$$D_S=\int\left(\,\cdot\,\mathcal{S}\right)\,dr=\int\mathrm{div}(\mathcal{E}\times\mathcal{H})\,dr=\text{dissipation engergy lost from radiation}\approx 0\text{ for low frequencies }\omega$$

$$d=\sqrt{\frac{1}{\mu\sigma}\frac{2}{\omega}}=\text{skin depth (surface) of conductor}\qquad r=\frac{d}{\sqrt{2i}}\rho=\frac{d}{1+i}\rho=\text{radius (extension) of conductor}$$

$$\rho=kr=\text{"radius of curvature }k\text{"}\qquad \omega=2\pi\nu=k\lambda\nu=\frac{1}{\sqrt{LC}}=\text{frequency}$$

$$\mu=\text{permeability of conductor}\qquad \sigma=\text{conductivity of conductor}$$

Low Frequency:

$$\omega\ll\frac{1}{\mu\sigma}\frac{2}{r^2}\iff r\ll d\iff\rho\ll 1\tag{12}$$

$$L \approx const. \qquad C \approx const. \qquad D_S \approx 0$$

High Frequency:

$$\omega \gg \frac{1}{\mu\sigma}\frac{2}{r^2} \iff r \gg d \iff \rho \gg 1 \tag{13}$$

$$L(t) \neq const. \qquad C(t) \neq const. \qquad \text{for Dipol: } D_S \propto \omega^4 \gg 0$$

Energy:

Action Quantum, Planck Constant (Energy-Frequency-Slope)[1]:

$$\begin{gathered} h = \tan(\alpha) = \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \\ \frac{E_1 - E_2}{\nu_0} = \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k \lambda \end{gathered} \tag{14}$$

$$\hbar = \frac{E_1 - E_2}{\omega} = \tag{15}$$

$$E_{kin}^{max} = eV_0 = e\frac{Q}{C} = h(\nu - \nu_0) \tag{16}$$

$$V_0 = \text{Opposing Potential } (I = 0)$$

$$\alpha = \angle(E_{kin}, \nu) \qquad e = \text{electron charge} \qquad Q = \text{ion charge}$$

$$C = \text{Capacitance} \qquad \nu = \frac{1}{2\pi}\frac{1}{\sqrt{LC}}$$

$$\Rightarrow \text{Inductance } L = \frac{4\pi^2}{C}\left(\frac{e}{h}Q + \nu_0\right)^2$$

[1]Wirkungsquantum

$\Rightarrow$ Ionisation Energy of chemical element